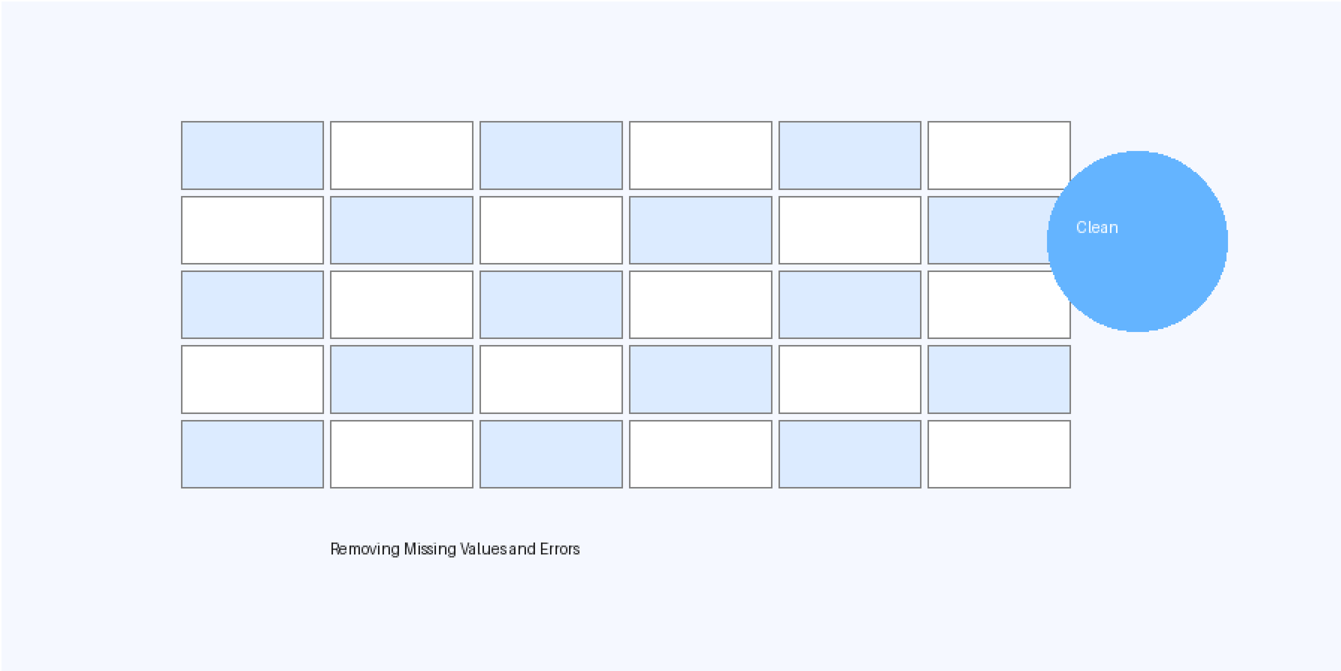
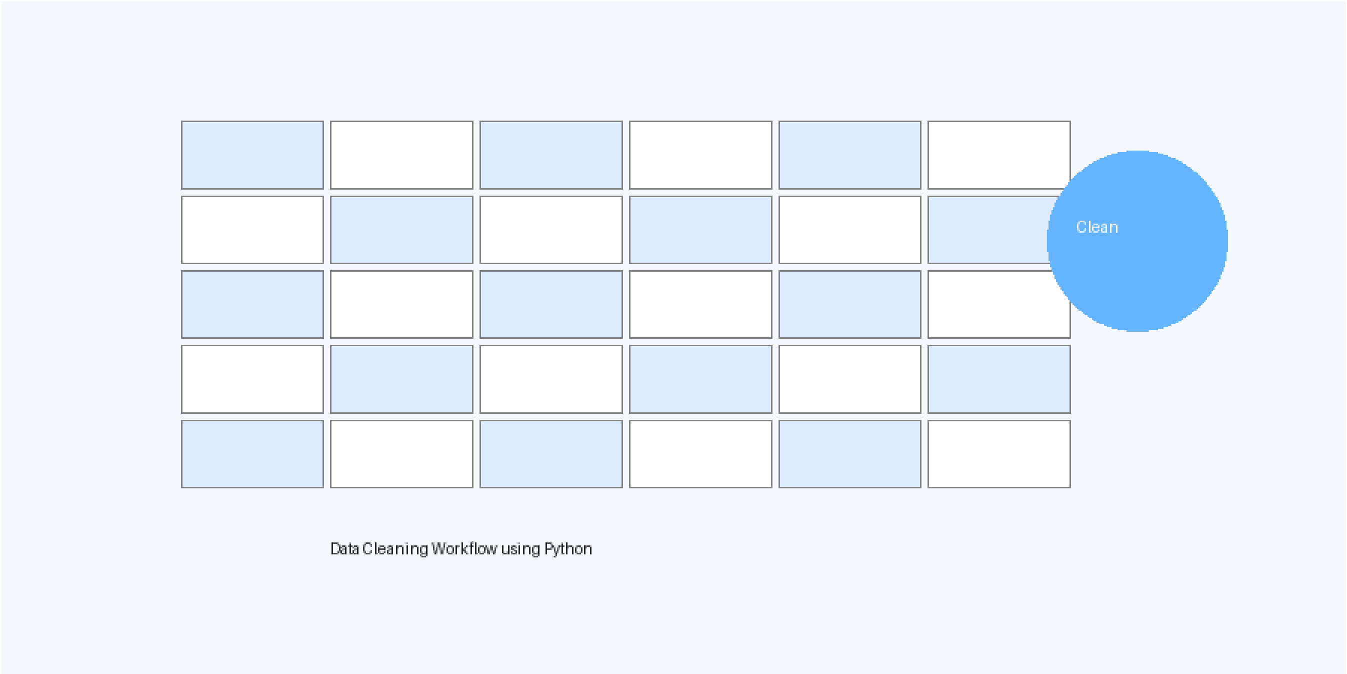
TASK 3 - DATA CLEANING PRACTICE

Learn Data Cleaning using Python and Pandas



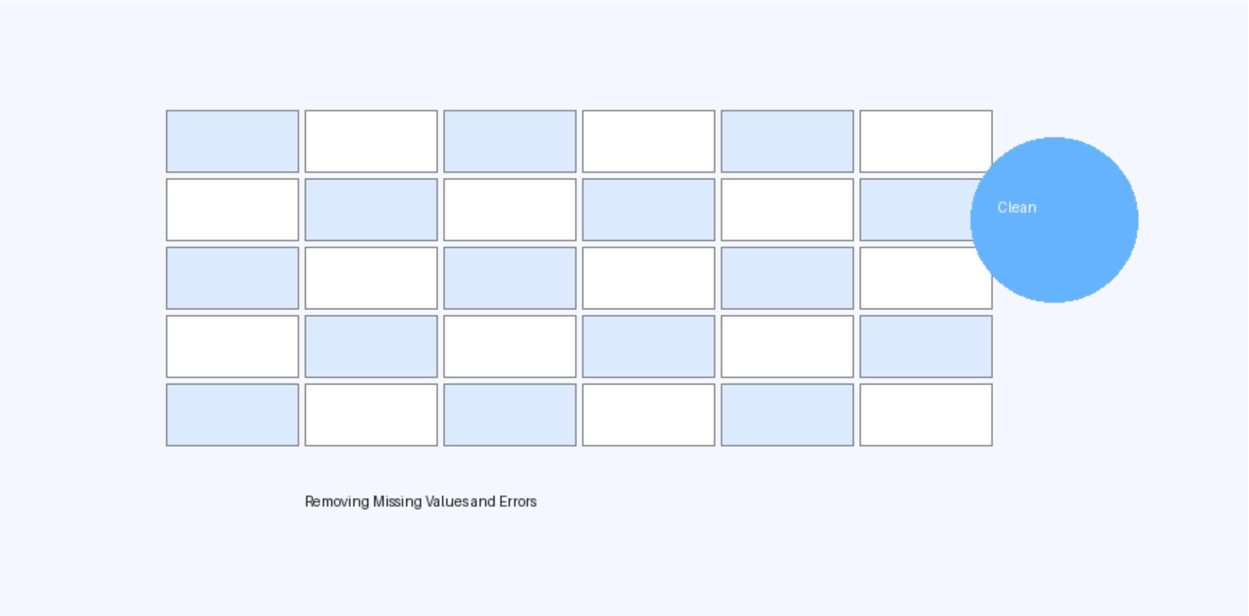
Project Objective

• Learn how to clean raw and messy datasets.
• Identify missing values and duplicate rows.
• Convert incorrect data types using Pandas.
• Save the cleaned dataset for analysis.



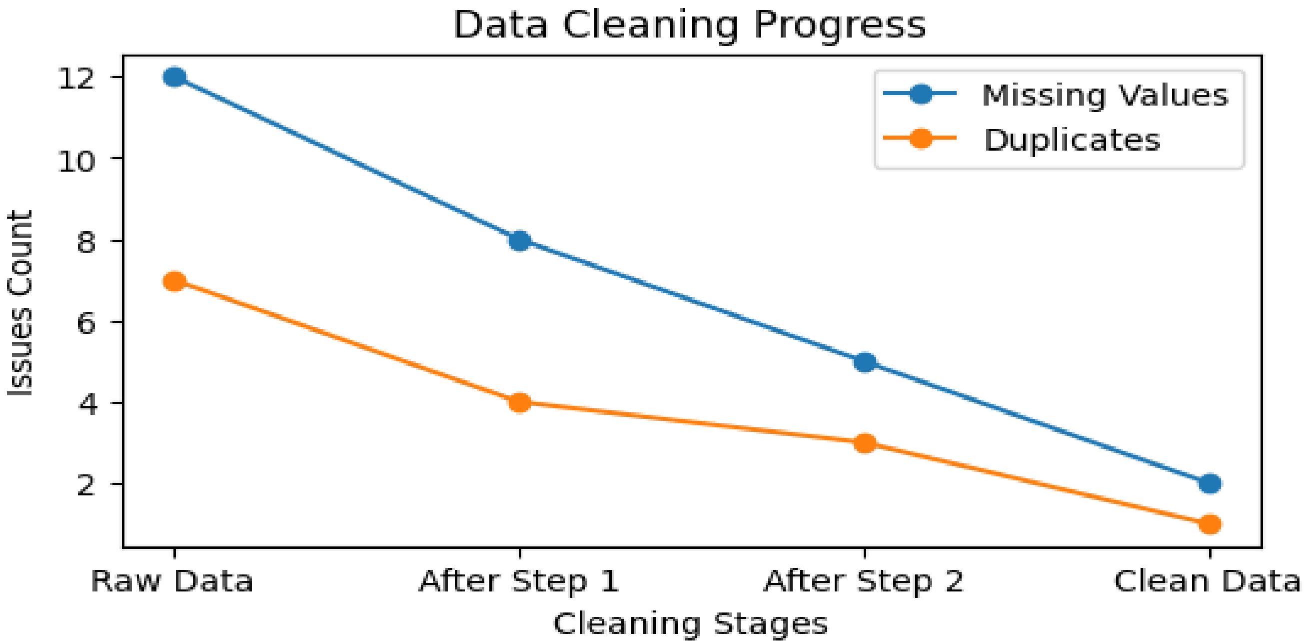
Data Cleaning Steps

1. Load dataset using **pd.read_csv()**.
2. Check missing values using **isnull().sum()**.
3. Remove or fill missing values.
4. Remove duplicate rows using **drop_duplicates()**.
5. Convert incorrect data types using **astype()**.



Data Cleaning Visualization

The graph below shows how data issues reduce after applying cleaning techniques.



Conclusion

• Data cleaning improves dataset quality and reliability.
• Clean data helps machine learning models perform better.
• Python and Pandas provide powerful tools for preprocessing.
• This task builds a strong foundation for data analytics projects.